Bayesian Decision Theory – Continuous Features

Chapter 02

Ref. Book: Pattern Recognition – S. Theodoridis

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Bayesian Classifier Examples

Example 01

Given:

- A doctor knows that meningitis causes stiff neck 50% of the time
- Prior probability of any patient having meningitis is 1/50000
- Prior probability of any patient having stiff neck is 1/20

Question: If a patient has stiff neck, does he/she has meningitis?

Bayesian Classifier Examples

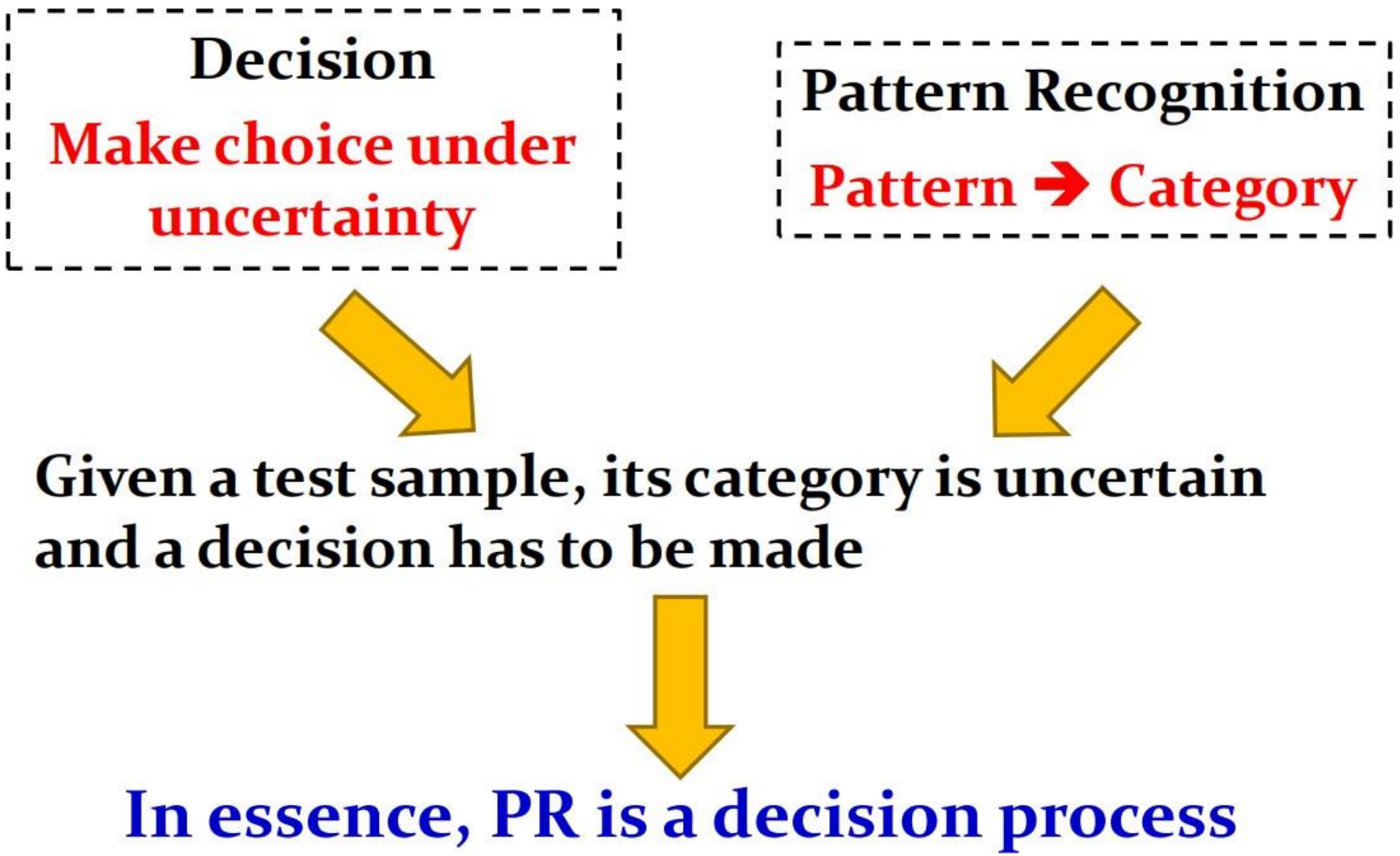
Example 02

The sea bass/salmon example

- We now the previous counts of salmon/sea bass
- Can we predict which fish is coming in the conveyor?

State of nature is random variable
The catch of salmon and sea bass is equiprobable
- P (ω_1) = P (ω_2) (uniform priors)
- P (ω_1) + P (ω_2) = 1 (exclusivity and exhaustivity)

Decision Theory



Bayesian Decision Theory

Bayesian decision theory is a statistical approach to pattern recognition

The fundamentals of most PR algorithms are rooted from Bayesian decision theory

Basic Assumptions

- The decision problem is posed (formalized) in **probabilistic** terms
- All the relevant probability values are known

Key Principle

Bayes Theorem

Bayes Theorem

Bayes theorem $$P(H|X) = \frac{P(H)P(X|H)}{P(X)}$$

X: the observed sample (also called **evidence**; *e.g.: the length of a fish*)

H: the hypothesis (*e.g. the fish belongs to the "salmon" category*)

$P(H)$: the **prior probability** (先验概率) that H holds (*e.g. the probability of catching a salmon*)

$P(X|H)$: the **likelihood** (似然度) of observing X given that H holds (e.g. *the probability of observing a 3-inch length fish which is salmon*)

$P(X)$: the **evidence probability** that X is observed (*e.g. the probability of observing a fish with 3-inch length*)

$P(H|X)$: the **posterior probability** (后验概率) that H holds given X (e.g. *the probability of X being salmon given its length is 3-inch*)



Thomas Bayes (1702-1761)

State of Nature

- Future events that might occur

  e.g. the next fish arriving along the conveyor belt

- State of nature is unpredictable

  e.g. it is hard to predict what type will emerge next



From statistical/probabilistic point of view, the state of nature should be favorably regarded as a random variable

e.g. let ω denote the (discrete) random variable representing the state of nature (class) of fish types

$\omega = \omega_1$: sea bass

$\omega = \omega_2$: salmon

Prior Probability

Prior probability is the probability distribution which reflects one's prior knowledge on the random variable

Probability distribution (for discrete random variable)

Let $P(\cdot)$ be the probability distribution on the random variable ω with c possible states of nature $\{\omega_1, \omega_2, \ldots, \omega_c\}$, such that:

$$P(\omega_i) \geq 0 \ (non\text{-}negativity) \qquad \sum_{i=1}^{c} P(\omega_i) = 1 \ (normalization)$$

the catch produced as much sea bass as salmon ⇒ $P(\omega_1) = P(\omega_2) = 1/2$

the catch produced more sea bass than salmon ⇒ $P(\omega_1) = 2/3; P(\omega_2) = 1/3$

Why Bayes?

- Decision rule with only the prior information
  - Decide ω_1 if P (ω_1)>P (ω_2), otherwise decide ω_2
  - Misclassify many fishes
- Use of the class-conditional information
  - Use lightness
- $P(x|\omega_1)$ and $P(x|\omega_2)$ describe lightness in salmon and seabass

Why Bayes (Cont.)

The Problem

To make a decision on the type of fish arriving next, where 1) prior probability is known; 2) no observation is allowed

Naive Decision Rule

Decide ω_1 if $P(\omega_1) > P(\omega_2)$; otherwise decide ω_2

- This is the *best* we can do without observation
- Fixed prior probabilities ➔ Same decisions all the time

Incorporate observations into decision!

Good when $P(\omega_1)$ *is much greater (smaller) than* $P(\omega_2)$

Poor when $P(\omega_1)$ *is close to* $P(\omega_2)$

[only 50% chance of being right if $P(\omega_1) = P(\omega_2)$]

Probability Density Function (PDF)

Probability density function (pdf)
(for continuous random variable)

Let $p(\cdot)$ be the probability density function on the continuous random variable x taking values in **R**, such that:

$$p(x) \geq 0 \ (non\text{-}negativity) \qquad \int_{-\infty}^{\infty} p(x)dx = 1 \ (normalization)$$

- For continuous random variable, it no longer makes sense to talk about the probability that *x* has a particular value (almost always be zero)
- We instead talk about the probability of *x* falling into a region *R*, say *R*=(*a*,*b*), which could be computed with the pdf:

$$\Pr[x \in R] = \int_{x \in R} p(x)dx = \int_{a}^{b} p(x)dx$$

Incorporate Observations

The Problem

Suppose the fish *lightness measurement x* is observed, how could we incorporate this knowledge into usage?

Class-conditional probability density function

- It is a probability density function (pdf) for x given that the state of nature (class) is ω , i.e.:

$p(x|\omega)$

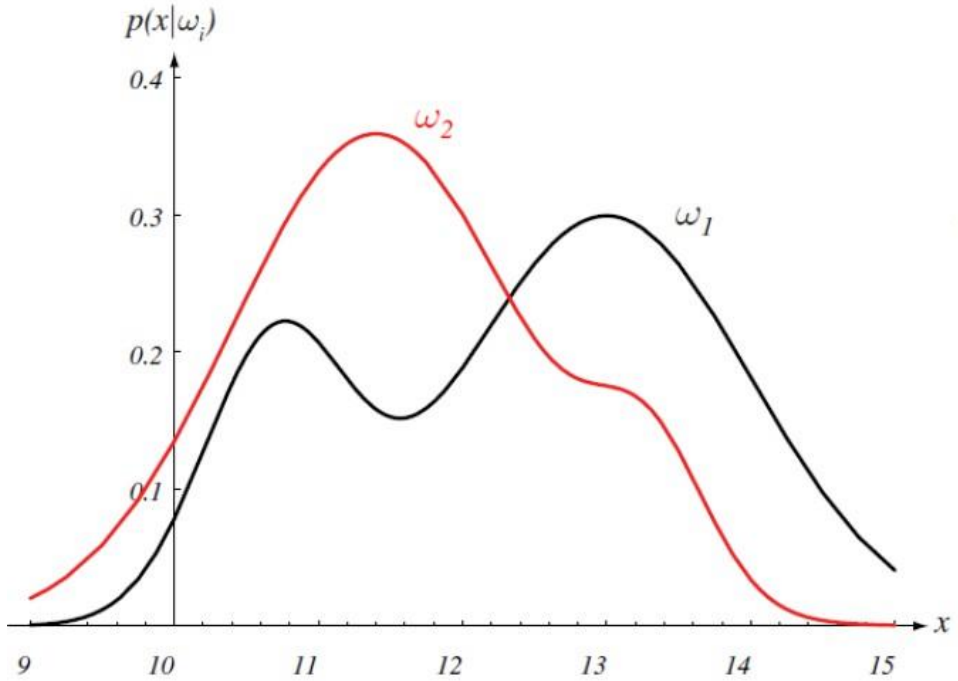
$$p(x|\omega) \geq 0 \qquad \int_{-\infty}^{\infty} p(x|\omega)dx = 1$$

- The *class-conditional* pdf describes the difference in the distribution of observations under different classes

$p(x|\omega_1)$ should be different to $p(x|\omega_2)$

Class Conditional PDF

An illustrative example



class-conditional pdf for *lightness*

h-axis: lightness of fish scales

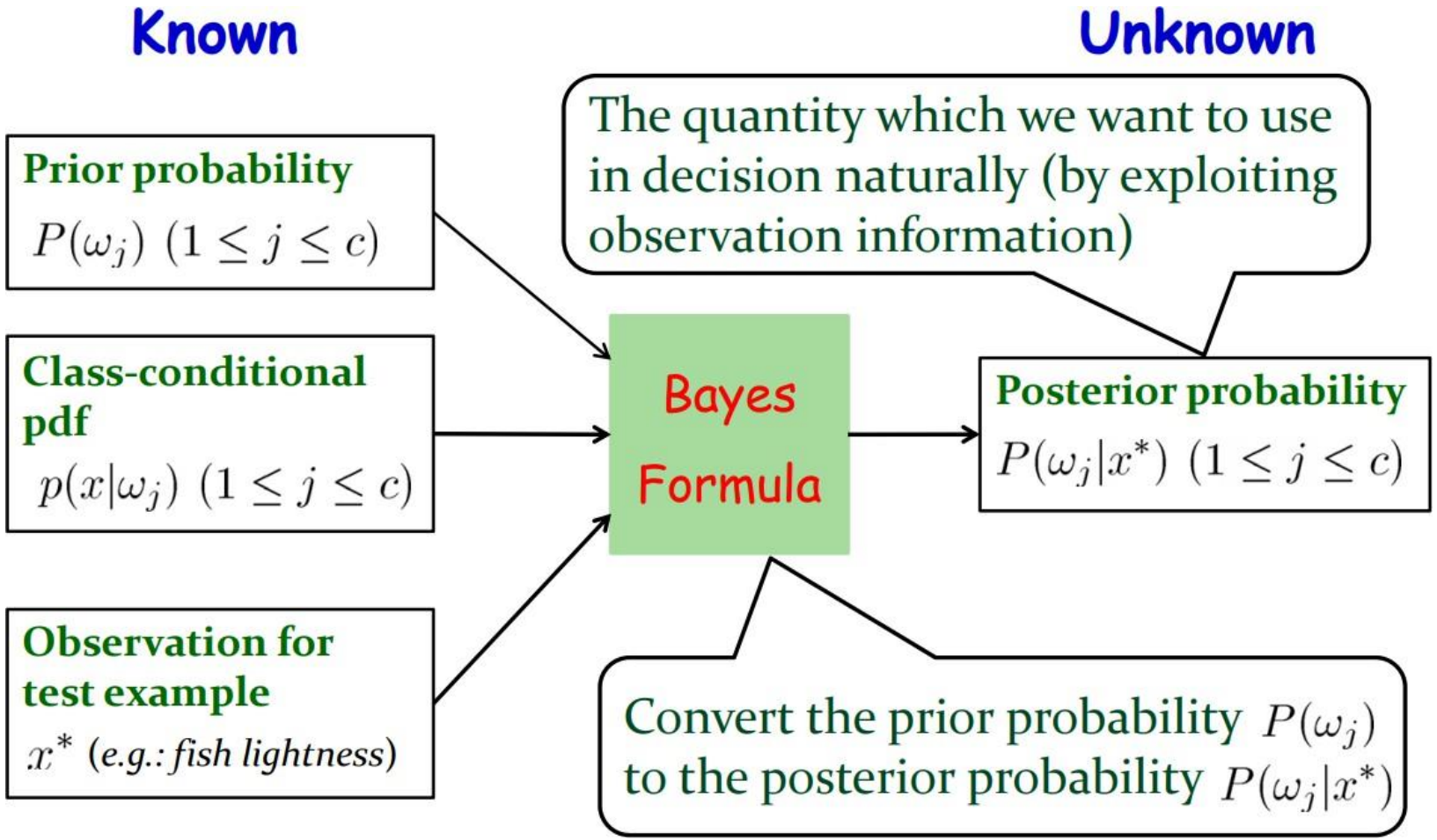
v-axis: class-conditional pdf values

black curve: sea bass

red curve: salmon

- The area under each curve is 1.0 (*normalization*)
- Sea bass is somewhat brighter than salmon

Decision After Observation



Bayes Formula Revisited

$$P(\omega_j|x) = \frac{p(x|\omega_j) \cdot P(\omega_j)}{p(x)} \quad (1 \leq j \leq c) \quad \text{(Bayes Formula)}$$

Bayes Decision Rule

$$\text{if } P(\omega_j|x) > P(\omega_i|x),\ \forall i \neq j \implies \text{Decide } \omega_j$$

- $P(\omega_j)$ and $p(x|\omega_j)$ are assumed to be known
- $p(x)$ is irrelevant for Bayesian decision (serving as a normalization factor, not related to any state of nature)

$$p(x) = \sum_{j=1}^{c} p(\omega_j, x) = \sum_{j=1}^{c} p(x|\omega_j) \cdot P(\omega_j)$$

Bayes Formula Revisited (Cont.)

$$P(\omega_j|x) = \frac{p(x|\omega_j) \cdot P(\omega_j)}{p(x)} \quad \left(posterior = \frac{likelihood \times prior}{evidence} \right)$$

Special Case I: Equal prior probability

$$P(\omega_1) = P(\omega_2) = \cdots = P(\omega_c) = \frac{1}{c}$$ ⇒ Depends on the likelihood $P(x|\omega_j)$

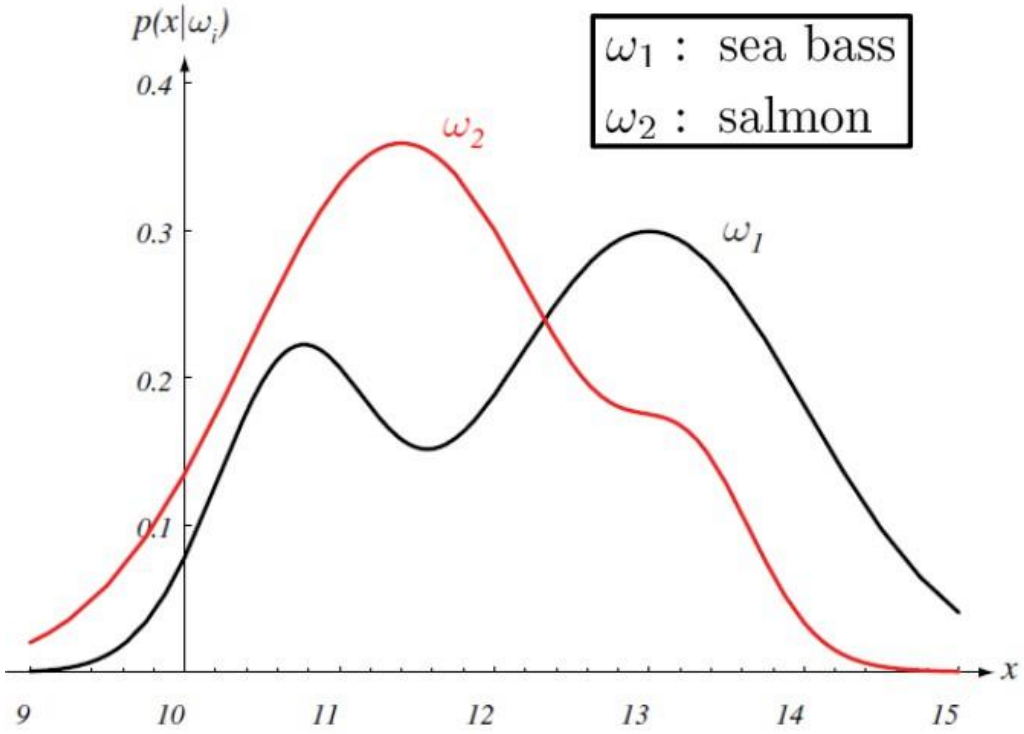
Special Case II: Equal likelihood

$$p(x|\omega_1) = p(x|\omega_2) = \cdots = p(x|\omega_c)$$ ⇒ Degenerate to naive decision rule

Normally, prior probability and likelihood function together in Bayesian decision process

Bayes Formula Revisited (Cont.)

An illustrative example



class-conditional pdf for *lightness*

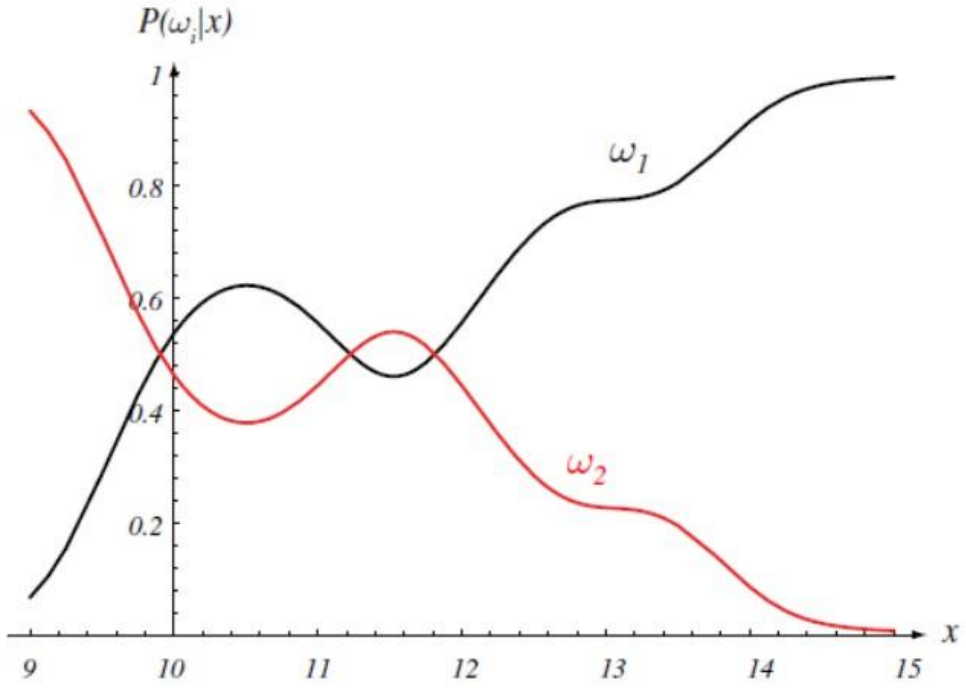
$$P(\omega_1) = \frac{2}{3}$$

$$P(\omega_2) = \frac{1}{3}$$

What will the posterior probability for either type of fish look like?

Bayes Formula Revisited (Cont.)

An illustrative example



posterior probability for either type of fish

h-axis: lightness of fish scales

v-axis: posterior probability for either type of fish

black curve: sea bass

red curve: salmon

- For each value of x, the higher curve yields the output of Bayesian decision
- For each value of x, the posteriors of either curve sum to 1.0

Another Example

Problem statement

- A new medical test is used to detect whether a patient has a certain cancer or not, whose test result is either + (*positive*) or - (*negative*)
- For patient with this cancer, the probability of returning *positive* test result is 0.98
- For patient without this cancer, the probability of returning *negative* test result is 0.97
- The probability for any person to have this cancer is 0.008

Question

If *positive* test result is returned for some person, does he/she have this kind of cancer or not?

Another Example (Cont.)

ω_1 : cancer $\qquad$ ω_2 : no cancer $\qquad$ $x \in \{+, -\}$

$$P(\omega_1) = 0.008 \qquad P(\omega_2) = 1 - P(\omega_1) = 0.992$$

$$P(+ \mid \omega_1) = 0.98 \qquad P(- \mid \omega_1) = 1 - P(+ \mid \omega_1) = 0.02$$

$$P(- \mid \omega_2) = 0.97 \qquad P(+ \mid \omega_2) = 1 - P(- \mid \omega_2) = 0.03$$

$$P(\omega_1 \mid +) = \frac{P(\omega_1)P(+ \mid \omega_1)}{P(+)} = \frac{P(\omega_1)P(+ \mid \omega_1)}{P(\omega_1)P(+ \mid \omega_1) + P(\omega_2)P(+ \mid \omega_2)}$$

$$= \frac{0.008 \times 0.98}{0.008 \times 0.98 + 0.992 \times 0.03} = 0.2085$$

$$P(\omega_2 \mid +) = 1 - P(\omega_1|+) = 0.7915$$

$P(\omega_2 \mid +) > P(\omega_1|+)$

No cancer!

Another Example (Cont.)

$$P(\omega|x) = \frac{p(x|\omega) \cdot P(\omega)}{p(x)} \quad \text{(Bayes Formula)}$$

To compute posterior probability $P(\omega|x)$, we need to know:

Prior probability: $P(\omega)$ Likelihood: $p(x|\omega)$

How do we know these probabilities?

- A simple solution: Counting

Relative Frequencies

- An advanced solution: Conduct

Density Estimation

Further Example

Problem statement

Based on the height of a car in some campus, decide whether it costs more than \$50,000 or not

ω_1 : price $> \$50,000$

ω_2 : price $\leq \$50,000$

x : height of car

$P(\omega_1|x) > P(\omega_2|x)$

?

$P(\omega_1|x) < P(\omega_2|x)$

Quantities to know:

$P(\omega_1)$ $P(\omega_2)$ $p(x|\omega_1)$ $p(x|\omega_2)$



Counting relative frequencies via collected samples

Further Example (Cont.)

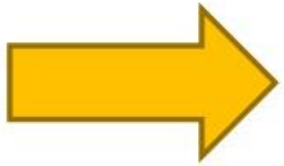
Collecting samples

Suppose we have randomly picked 1209 cars in the campus, got prices from their owners, and measured their heights

Compute $P(\omega_1), P(\omega_2)$:

cars in ω_1 : 221

cars in ω_2 : 988



$$P(\omega_1) = \frac{221}{1209} = 0.183$$

$$P(\omega_2) = \frac{988}{1209} = 0.817$$

Further Example (Cont.)

Compute $p(x|\omega_1), p(x|\omega_2)$:

Discretize the height spectrum (say [0.5m, 2.5m]) into 20 intervals each with length 0.1m, and then count the number of cars falling into each interval for either class

Suppose $x = 1.05$

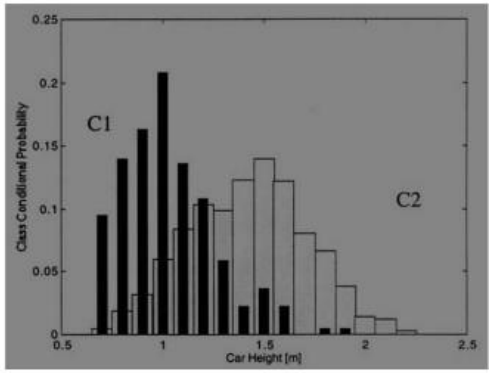
x falls into interval I_x=[1.0m, 1.1m]

For ω_1, # cars in I_x is 46

For ω_2, # cars in I_x is 59

$$p(x = 1.05|\omega_1) = \frac{46}{221} = 0.2081$$

$$p(x = 1.05|\omega_2) = \frac{59}{988} = 0.0597$$



Further Example (Cont.)

Question

For a car with height 1.05m, is its price greater than \$50,000?

Estimated quantities

$P(\omega_1) = 0.183$ $\quad P(\omega_2) = 0.817$

$p(x = 1.05 \mid \omega_1) = 0.2081$ $\quad p(x = 1.05 \mid \omega_2) = 0.0597$

$$\frac{P(\omega_2 \mid x = 1.05)}{P(\omega_1 \mid x = 1.05)} = \frac{P(\omega_2) \cdot p(x = 1.05 \mid \omega_2)}{p(x = 1.05)} \Big/ \frac{P(\omega_1) \cdot p(x = 1.05 \mid \omega_1)}{p(x = 1.05)}$$

$$= \frac{P(\omega_2) \cdot p(x = 1.05 \mid \omega_2)}{P(\omega_1) \cdot p(x = 1.05 \mid \omega_1)}$$

$$= \frac{0.817 \times 0.0597}{0.183 \times 0.2081} = 1.280$$

$P(\omega_2 \mid x) > P(\omega_1 \mid x)$

$\text{price} \leq \$50,000$

Is Bayes Decision Rule Optimal?

Bayes Decision Rule (In case of two classes)

if $P(\omega_1|x) > P(\omega_2|x)$, Decide ω_1; Otherwise ω_2

Whenever we observe a particular x, the probability of error is:

$$P(error \mid x) = \begin{cases} P(\omega_1 \mid x) & \text{if we decide } \omega_2 \\ P(\omega_2 \mid x) & \text{if we decide } \omega_1 \end{cases}$$

Under Bayes decision rule, we have

$$P(error \mid x) = \min[P(\omega_1 \mid x), P(\omega_2 \mid x)]$$

For every x, we ensure that $P(error \mid x)$ is as small as possible



The average probability of error over all possible x must be as small as possible

Bayes Decision Rules

- Generalization of the preceding ideas
  - Use of more than one feature
  - Use more than two states of nature
  - Allowing actions and not only decide on the state of nature
  - Introduce a loss function which is more general than the probability of error

The General Case

- By allowing to use more than one feature

  $x \in \mathbf{R} \implies \mathbf{x} \in \mathbf{R}^d$ (d-dimensional Euclidean space)

- By allowing more than two states of nature

  $\Omega = \{\omega_1, \omega_2, \ldots, \omega_c\}$ (finite set of c states of nature)

- By allowing actions other than merely deciding the state of nature

  $\mathcal{A} = \{\alpha_1, \alpha_2, \ldots, \alpha_a\}$ (finite set of a possible actions)

Note : $c \neq a$

The General Case (Cont.)

- By introducing a loss function more general than the probability of error

$$\lambda : \ \Omega \times \mathcal{A} \rightarrow \mathbf{R} \quad \text{(loss function)}$$

$\lambda(\omega_j, \alpha_i)$: the loss incurred for taking action α_i when the state of nature is ω_j

For ease of reference, usually written as:

$$\lambda(\alpha_i \mid \omega_j)$$

A simple loss function

Class \ Action	$\alpha_1 =$ "Recipe A"	$\alpha_2 =$ "Recipe B"	$\alpha_3 =$ "No Recipe"
$\omega_1 =$ "cancer"	**5**	**50**	**10,000**
$\omega_2 =$ "no cancer"	**60**	**3**	**0**

The General Case (Cont.)

The problem

Given a particular **x**, we have to decide which action to take



We need to know the *loss* of taking each action α_i $(1 \leq i \leq a)$

true state of nature is ω_j

the action being taken is α_i



incur the loss $\lambda(\alpha_i \mid \omega_j)$

However, the true state of nature is uncertain



Expected (average) loss

The General Case (Cont.)

Expected Loss

$$R(\alpha_i \mid \mathbf{x}) = \sum_{j=1}^{c} \lambda(\alpha_i \mid \omega_j) \cdot P(\omega_j \mid \mathbf{x})$$

Average by enumerating over all possible states of nature

The incurred loss of taking action α_i in case of true state of nature being ω_j

The probability of ω_j being the true state of nature

The expected loss is also named as ***(conditional)***

Risk

The General Case (Cont.)

Suppose we have:

Class \ Action	α_1 = "Recipe A"	α_2 = "Recipe B"	α_3 = "No Recipe"
ω_1 = "cancer"	**5**	**50**	**10,000**
ω_2 = "no cancer"	**60**	**3**	**0**

For a particular **x**:

$$P(\omega_1 \mid \mathbf{x}) = 0.01$$

$$P(\omega_2 \mid \mathbf{x}) = 0.99$$

$$R(\alpha_1 \mid \mathbf{x}) = \sum_{j=1}^{2} \lambda(\alpha_1 \mid \omega_j) \cdot P(\omega_j \mid \mathbf{x})$$

$$= \lambda(\alpha_1 \mid \omega_1) \cdot P(\omega_1 \mid \mathbf{x}) + \lambda(\alpha_1 \mid \omega_2) \cdot P(\omega_2 \mid \mathbf{x})$$

$$= 5 \times 0.01 + 60 \times 0.99 = 59.45$$

Similarly, we can get: $R(\alpha_2 \mid \mathbf{x}) = 3.47 \quad R(\alpha_3 \mid \mathbf{x}) = 100$

The General Case (Cont.)

The task: *find a mapping from patterns to actions*

$$\alpha : \mathbf{R}^d \rightarrow \mathcal{A} \quad \text{(decision function)}$$

In other words, for every $\mathbf{x}$, the decision function $\alpha(\mathbf{x})$ assumes one of the a actions $\alpha_1, \ldots, \alpha_a$

Overall risk R

expected loss with decision function $\alpha(\cdot)$

$$R = \int \underline{\underline{R(\alpha(\mathbf{x}) \mid \mathbf{x})}} \cdot \underline{\underline{p(\mathbf{x})}} d\mathbf{x}$$

Conditional risk for pattern $\mathbf{x}$ with action $\alpha(\mathbf{x})$

pdf for patterns

The General Case (Cont.)

$$R = \int R(\alpha(\mathbf{x}) \mid \mathbf{x})) \cdot p(\mathbf{x}) d\mathbf{x} \quad \text{(overall risk)}$$

For every **x**, we ensure that the conditional risk $R(\alpha(\mathbf{x}) \mid \mathbf{x})$ is as small as possible



The overall risk over all possible **x** must be as small as possible

Bayes decision rule (*General case*)

$$\alpha(\mathbf{x}) = \arg\min_{\alpha_i \in \mathcal{A}} R(\alpha_i \mid \mathbf{x})$$

$$= \arg\min_{\alpha_i \in \mathcal{A}} \sum_{j=1}^{c} \lambda(\alpha_i \mid \omega_j) \cdot P(\omega_j \mid \mathbf{x})$$

- The resulting overall risk is called the ***Bayes risk*** (denoted as R^*)
- The best performance achievable given $p(\mathbf{x})$ and loss function

Two Category Classification

Special case

λ_{11} λ_{21}

λ_{12} λ_{22}

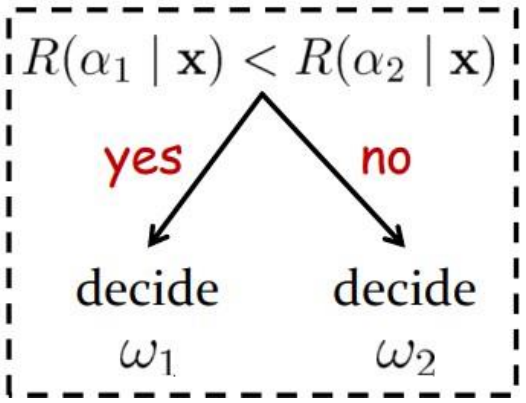
- $\Omega = \{\omega_1, \omega_2\}$ (two states of nature)
- $\mathcal{A} = \{\alpha_1, \alpha_2\}$ ($\alpha_1 =$ decide ω_1; $\alpha_2 =$ decide ω_2)

$\lambda_{ij} = \lambda(\alpha_i \mid \omega_j)$: the loss incurred for deciding ω_i when the true state of nature is ω_j

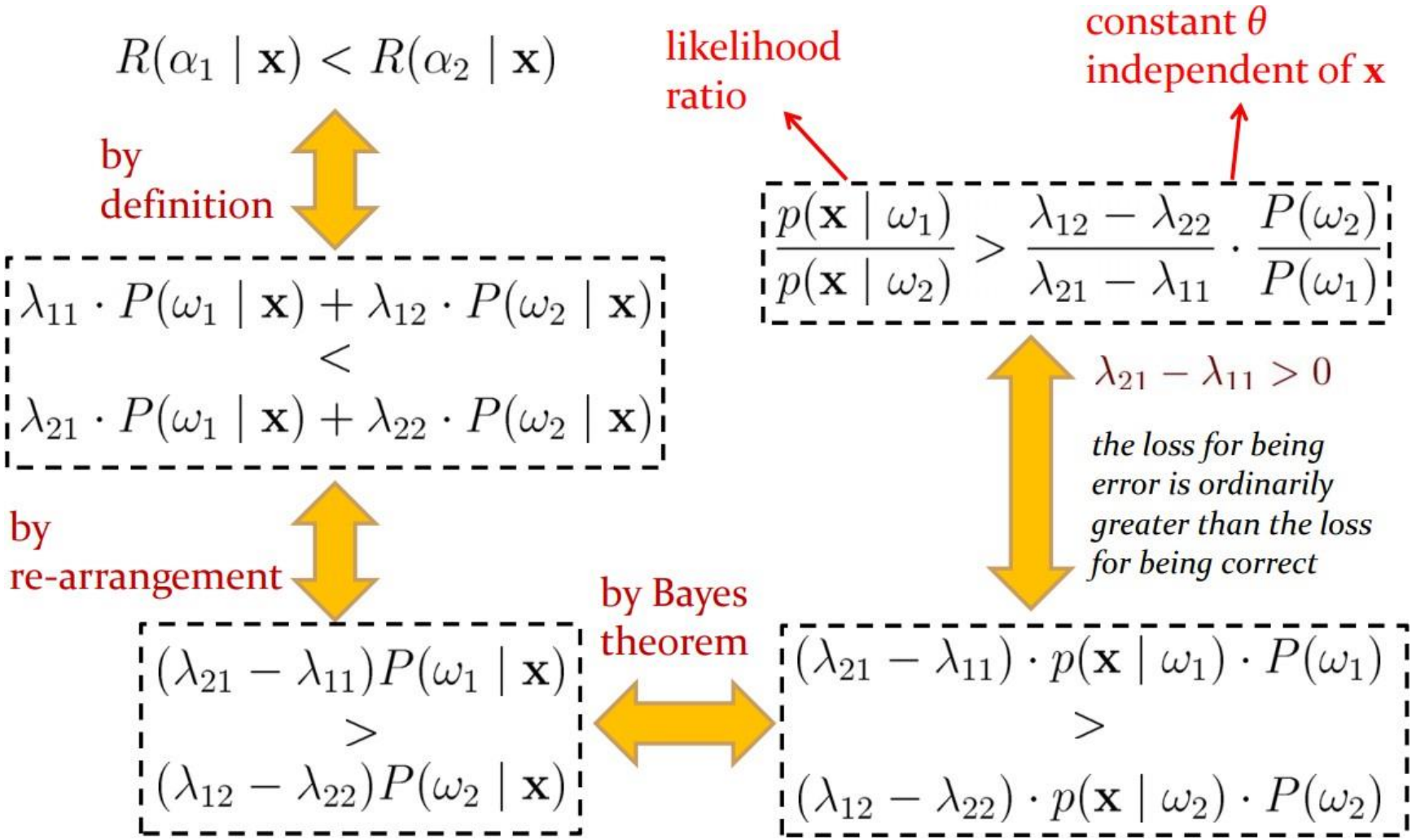
The conditional risk:

$$R(\alpha_1 \mid \mathbf{x}) = \lambda_{11} \cdot P(\omega_1 \mid \mathbf{x}) + \lambda_{12} \cdot P(\omega_2 \mid \mathbf{x})$$

$$R(\alpha_2 \mid \mathbf{x}) = \lambda_{21} \cdot P(\omega_1 \mid \mathbf{x}) + \lambda_{22} \cdot P(\omega_2 \mid \mathbf{x})$$



Two Category Classification (Cont.)



Minimum Error-rate Classification

Classification setting

- $\Omega = \{\omega_1, \omega_2, \ldots, \omega_c\}$ (c possible states of nature)
- $\mathcal{A} = \{\alpha_1, \alpha_2, \ldots, \alpha_c\}$ (α_i = decide ω_i, $1 \leq i \leq c$)

Zero-one (symmetrical) loss function

$$\lambda(\alpha_i \mid \omega_j) = \begin{cases} 0 & i = j \\ 1 & i \neq j \end{cases} \qquad 1 \leq i, j \leq c$$

- Assign no loss (i.e. 0) to a correct decision
- Assign a unit loss (i.e. 1) to any incorrect decision (**equal cost**)

Minimum Error-rate Classification (Cont.)

$$R(\alpha_i \mid \mathbf{x}) = \sum_{j=1}^{c} \lambda(\alpha_i \mid \omega_j) \cdot P(\omega_j \mid \mathbf{x})$$

$$= \sum_{j \neq i} \lambda(\alpha_i \mid \omega_j) \cdot P(\omega_j \mid \mathbf{x}) \; + \; \lambda(\alpha_i \mid \omega_i) \cdot P(\omega_i \mid \mathbf{x})$$

$$= \sum_{j \neq i} P(\omega_j \mid \mathbf{x})$$

$$= 1 - P(\omega_i \mid \mathbf{x})$$

Error Rate

the probability that action α_i (decide ω_i) is wrong

Minimum error rate

Decide ω_i if $P(\omega_i \mid \mathbf{x}) > P(\omega_j \mid \mathbf{x})$ for all $j \neq i$

Discriminant Function

Classification

Pattern → Category

actions ⟷ **decide categories**

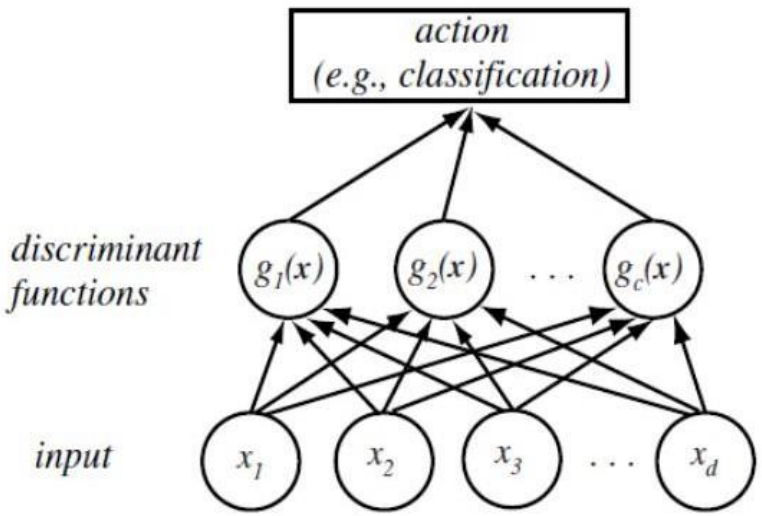
Discriminant functions

$g_i : \mathbf{R}^d \rightarrow \mathbf{R} \quad (1 \leq i \leq c)$

- *Useful way to represent classifiers*
- *One function per category*

Decide ω_i

if $g_i(\mathbf{x}) > g_j(\mathbf{x})$ for all $j \neq i$



Discriminant Function (Cont.)

Minimum risk: $g_i(\mathbf{x}) = -R(\alpha_i \mid \mathbf{x}) \quad (1 \le i \le c)$

Minimum-error-rate: $g_i(\mathbf{x}) = P(\omega_i|\mathbf{x}) \quad (1 \le i \le c)$

Various discriminant functions ⇒ **Identical classification results**

$f(\cdot)$ is a *monotonically increasing function*

⇒ $f(g_i(\mathbf{x})) \Longleftrightarrow g_i(\mathbf{x})$ *(i.e. equivalent in decision)*

e.g.:

$f(x) = k \cdot x \ (k > 0)$ ⇒ $f(g_i(\mathbf{x})) = k \cdot g_i(\mathbf{x}) \ (1 \le i \le c)$

$f(x) = \ln x$ ⇒ $f(g_i(\mathbf{x})) = \ln g_i(\mathbf{x}) \ (1 \le i \le c)$

Discriminant Function (Cont.)

Decision region

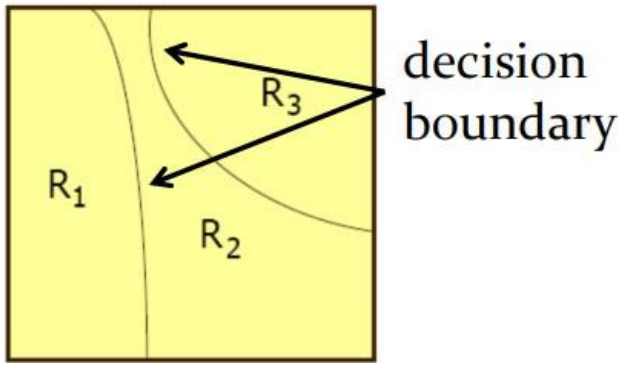
c discriminant functions $g_i(\cdot)\ (1 \le i \le c)$ → c decision regions $\mathcal{R}_i \subset \mathbf{R}^d\ (1 \le i \le c)$

$$\mathcal{R}_i = \{\mathbf{x} \mid \mathbf{x} \in \mathbf{R}^d : g_i(\mathbf{x}) > g_j(\mathbf{x}) \quad \forall j \neq i\}$$

where $\mathcal{R}_i \cap \mathcal{R}_j = \emptyset\ (i \neq j)$ and $\bigcup_{i=1}^{c} \mathcal{R}_i = \mathbf{R}^d$

Decision boundary

surface in feature space where ties occur among several largest discriminant functions



Expected Value

Expected Value, a.k.a. *expectation*, *mean* or *average* of a random variable x

Discrete case

$x \in \mathcal{X} = \{x_1, x_2, \ldots, x_c\}$

$x \sim P(\cdot)$

($\sim$: "has the distribution")

$$\mathcal{E}[x] = \sum_{x \in \mathcal{X}} x \cdot P(x) = \sum_{i=1}^{c} x_i \cdot P(x_i)$$

Continuous case

Notation: $\mu = \mathcal{E}[x]$

$x \in \mathbf{R}$

$x \sim p(\cdot)$

$$\mathcal{E}[x] = \int_{-\infty}^{\infty} x \cdot p(x)\, dx$$

Expected Value



Given random variable x and function $f(\cdot)$, what is the expected value of $f(x)$?

Discrete case: $\mathcal{E}[f(x)] = \sum_{x \in \mathcal{X}} f(x) \cdot P(x) = \sum_{i=1}^{c} f(x_i) \cdot P(x_i)$

Continuous case: $\mathcal{E}[f(x)] = \int_{-\infty}^{\infty} f(x) \cdot p(x)\, dx$

Variance $\mathrm{Var}[x] = \mathcal{E}[(x - \mathcal{E}[x])^2]$ (i.e. $f(x) = (x - \mu)^2$)

Discrete case: $\mathrm{Var}[x] = \sum_{i=1}^{c} (x_i - \mu)^2 \cdot P(x_i)$

Continuous case: $\mathrm{Var}[x] = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot p(x)\, dx$

Notation: $\sigma^2 = \mathrm{Var}[x]$ (σ: *standard deviation*)

Gaussian Density- Univariate Case

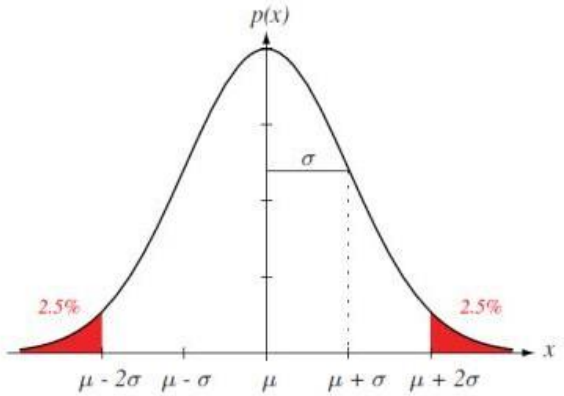
Gaussian density , a.k.a. *normal density*

for continuous random variable

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \qquad x \sim N(\mu, \sigma^2)$$

$$\int_{-\infty}^{\infty} p(x)dx = 1$$

$$\mathcal{E}[x] = \int_{-\infty}^{\infty} x \cdot p(x) = \mu$$



$$\mathrm{Var}[x] = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot p(x) = \sigma^2$$

Vector Random Variable

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix}$$

$$\mathbf{x} \sim p(\mathbf{x}) = p(x_1, x_2, \ldots, x_d) \quad \textbf{(joint pdf)}$$

$$p(\mathbf{x}_1) = \int p(\mathbf{x}_1, \mathbf{x}_2)\, d\mathbf{x}_2 \quad \textbf{(marginal pdf)}$$

$$(\mathbf{x}_1 \cap \mathbf{x}_2 = \emptyset;\ \mathbf{x}_1 \cup \mathbf{x}_2 = \mathbf{x})$$

Expected vector

$$\mathcal{E}[\mathbf{x}] = \begin{pmatrix} \mathcal{E}[x_1] \\ \mathcal{E}[x_2] \\ \vdots \\ \mathcal{E}[x_d] \end{pmatrix}$$

$$\mathcal{E}[x_i] = \int_{-\infty}^{\infty} x_i \cdot p(x_i)\, dx_i \quad (1 \le i \le d)$$

marginal pdf on the *i*-th component

Notation:

$$\boldsymbol{\mu} = \mathcal{E}[\mathbf{x}];\ \mu_i = \mathcal{E}[x_i]\ (1 \le i \le d)$$

Vector Random Variable

Covariance matrix

$$\boldsymbol{\Sigma} = [\sigma_{ij}]_{1 \le i,j \le d} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_{dd} \end{pmatrix}$$

Properties of $\boldsymbol{\Sigma}$

- symmetric
- Positive semidefinite

$$\sigma_{ij} = \sigma_{ji} = \mathcal{E}[(x_i - \mu_i)(x_j - \mu_j)]$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x_i - \mu_i)(x_j - \mu_j) \cdot p(x_i, x_j)\, dx_i dx_j$$

$$\sigma_{ii} = \mathrm{Var}[x_i] = \sigma_i^2$$

marginal pdf on a pair of random variables (x_i, x_j)

Gaussian Density- Multivariate Case

$$\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \qquad \mu_i = \mathcal{E}[x_i] \quad \sigma_{ij} = \sigma_{ji} = \mathcal{E}[(x_i - \mu_i)(x_j - \mu_j)]$$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right]$$

$\mathbf{x} = (x_1, x_2, \ldots, x_d)^t$: d-dimensional *column vector*

$\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_d)^t$: d-dimensional *mean vector*

$$\boldsymbol{\Sigma} = [\sigma_{ij}]_{1 \le i,j \le d} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_{dd} \end{pmatrix}$$

$d \times d$ *covariance matrix*

$|\boldsymbol{\Sigma}|$: determinant

$\boldsymbol{\Sigma}^{-1}$: inverse

Gaussian Density- Multivariate Case

$$\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}): \quad p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right]$$

$(\mathbf{x}-\boldsymbol{\mu})^t$: $1 \times d$ matrix

$\boldsymbol{\Sigma}^{-1}$: $d \times d$ matrix

$(\mathbf{x}-\boldsymbol{\mu})$: $d \times 1$ matrix



$(\mathbf{x}-\boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})$

scalar $(1 \times 1$ matrix$)$

$\boldsymbol{\Sigma}$: positive definite



$\boldsymbol{\Sigma}^{-1}$: positive definite



$(\mathbf{x}-\boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \geq 0$



$-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \leq 0$

Discriminant Function for Gaussian Density

Minimum-error-rate classification

$$g_i(\mathbf{x}) = P(\omega_i|\mathbf{x}) \quad (1 \le i \le c)$$

$$g_i(\mathbf{x}) = P(\omega_i|\mathbf{x}) \iff g_i(\mathbf{x}) = \ln P(\omega_i|\mathbf{x})$$

$$g_i(\mathbf{x}) = \ln p(\mathbf{x}|\omega_i) + \ln P(\omega_i)$$

$$p(\mathbf{x}|\omega_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^t \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\boldsymbol{\Sigma}_i| + \ln P(\omega_i)$$

Constant, could be ignored

Discriminant Function for Gaussian Density (Cont.)

Case I: $\boldsymbol{\Sigma}_i = \sigma^2 \mathbf{I}$

$$p(\mathbf{x}|\omega_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^t \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i) - \frac{1}{2}\ln|\boldsymbol{\Sigma}_i| + \ln P(\omega_i)$$

Covariance matrix: σ^2 times the identity matrix $\mathbf{I}$

$$\boldsymbol{\Sigma}_i = \sigma^2 \cdot \begin{pmatrix} 1 & & \\ & 1 & \\ & & \ddots & \\ & & & 1 \end{pmatrix} = \begin{pmatrix} \sigma^2 & & \\ & \sigma^2 & \\ & & \ddots & \\ & & & \sigma^2 \end{pmatrix} \Rightarrow \quad |\boldsymbol{\Sigma}_i| = \sigma^{2d}$$

$$\boldsymbol{\Sigma}_i^{-1} = (1/\sigma^2)\,\mathbf{I}$$

$$g_i(\mathbf{x}) = -\frac{||\mathbf{x}-\boldsymbol{\mu}_i||^2}{2\sigma^2} + \ln P(\omega_i)$$

$||\cdot||$: *Euclidean norm*

$$||\mathbf{x}-\boldsymbol{\mu}_i||^2 = (\mathbf{x}-\boldsymbol{\mu}_i)^t(\mathbf{x}-\boldsymbol{\mu}_i)$$

Discriminant Function for Gaussian Density (Cont.)

Case I: $\Sigma_i = \sigma^2 \mathbf{I}$ (Cont.)

$$g_i(\mathbf{x}) = -\frac{||\mathbf{x} - \boldsymbol{\mu}_i||^2}{2\sigma^2} + \ln P(\omega_i)$$



$$g_i(\mathbf{x}) = -\frac{1}{2\sigma^2}[\mathbf{x}^t\mathbf{x} - 2\boldsymbol{\mu}_i^t\mathbf{x} + \boldsymbol{\mu}_i^t\boldsymbol{\mu}_i] + \ln P(\omega_i)$$

the same for all *states of nature*, could be ignored

Linear discriminant functions

$$g_i(\mathbf{x}) = \mathbf{w}_i^t\,\mathbf{x} + w_{i0}$$

$$\mathbf{w}_i = \frac{1}{\sigma^2}\boldsymbol{\mu}_i \quad \textit{weight vector}$$

$$w_{i0} = -\frac{1}{2\sigma^2}\boldsymbol{\mu}_i^t\boldsymbol{\mu}_i + \ln P(\omega_i) \quad \textit{threshold/bias}$$

Discriminant Function for Gaussian Density (Cont.)

Case II: $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}$

$$p(\mathbf{x}|\omega_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^t \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i) - \frac{1}{2}\ln|\boldsymbol{\Sigma}_i| + \ln P(\omega_i)$$

Covariance matrix: *identical* for all classes

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^t \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_i) + \ln P(\omega_i)$$

$(\mathbf{x}-\boldsymbol{\mu}_i)^t \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu}_i)$: squared *Mahalanobis distance* (马氏距离)

$\boldsymbol{\Sigma} = \mathbf{I}$ ⇒ reduces to *Euclidean distance*



P. C. Mahalanobis (1893-1972)

Discriminant Function for Gaussian Density (Cont.)

Case II: $\Sigma_i = \Sigma$ (Cont.)

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^t\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_i) + \ln P(\omega_i)$$



$$g_i(\mathbf{x}) = -\frac{1}{2}[\mathbf{x}^t\boldsymbol{\Sigma}^{-1}\mathbf{x} - 2\boldsymbol{\mu}_i^t\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_i^t\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_i] + \ln P(\omega_i)$$

the same for all *states of nature*, could be ignored

Linear discriminant functions

$$g_i(\mathbf{x}) = \mathbf{w}_i^t\,\mathbf{x} + w_{i0}$$

$\mathbf{w}_i = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_i$ *weight vector*

$w_{i0} = -\frac{1}{2}\boldsymbol{\mu}_i^t\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_i + \ln P(\omega_i)$ *threshold/bias*

Discriminant Function for Gaussian Density (Cont.)

Case III: $\boldsymbol{\Sigma}_i = \text{arbitrary}$

$p(\mathbf{x}|\omega_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^t \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{1}{2}\ln|\boldsymbol{\Sigma}_i| + \ln P(\omega_i)$$

quadratic discriminant functions (二次判别函数)

$$g_i(\mathbf{x}) = \mathbf{x}^t \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^t \mathbf{x} + w_{i0}$$

$\mathbf{W}_i = -\frac{1}{2}\boldsymbol{\Sigma}_i^{-1}$ *quadratic matrix*

$\mathbf{w}_i = \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i$ *weight vector*

$w_{i0} = -\frac{1}{2}\boldsymbol{\mu}_i^t \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \frac{1}{2}\ln|\boldsymbol{\Sigma}_i| + \ln P(\omega_i)$ *threshold/bias*

The End